

Data with Pandas

Part of **Data Science Starter** · Estimated time: 50 min

Introduction

pandas is the standard Python library for working with tabular data — the kind you'd otherwise put in a spreadsheet. Its core object, the DataFrame, represents a table with labeled rows and columns.

Core ideas

- `pd.read_csv("file.csv")` loads a spreadsheet into a DataFrame in one line.
- Filtering rows uses a boolean condition: `df[df["age"] > 30]`. Selecting columns uses their names: `df[["name", "age"]]`.
- Missing values show up as NaN. `df.dropna()` removes rows with any missing values; `df.fillna(0)` replaces them with a chosen value — which one you pick depends on what the missing value actually means.
- `df.groupby("category").mean()` is one of the most common patterns: split the data into groups, then summarize each group.

Example

```
import pandas as pd
df = pd.read_csv("sales.csv")
df = df.dropna(subset=["revenue"])
print(df.groupby("region")["revenue"].sum())
```

Try it yourself

Find any small CSV file (or make one with 10-20 rows in a spreadsheet and export it), load it with pandas, and print the average of one numeric column grouped by one categorical column.

Further reading

- [pandas — Getting Started](https://pandas.pydata.org/docs/getting_started/index.html)

https://pandas.pydata.org/docs/getting_started/index.html

- [pandas — 10 Minutes to pandas](https://pandas.pydata.org/docs/user_guide/10min.html)

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